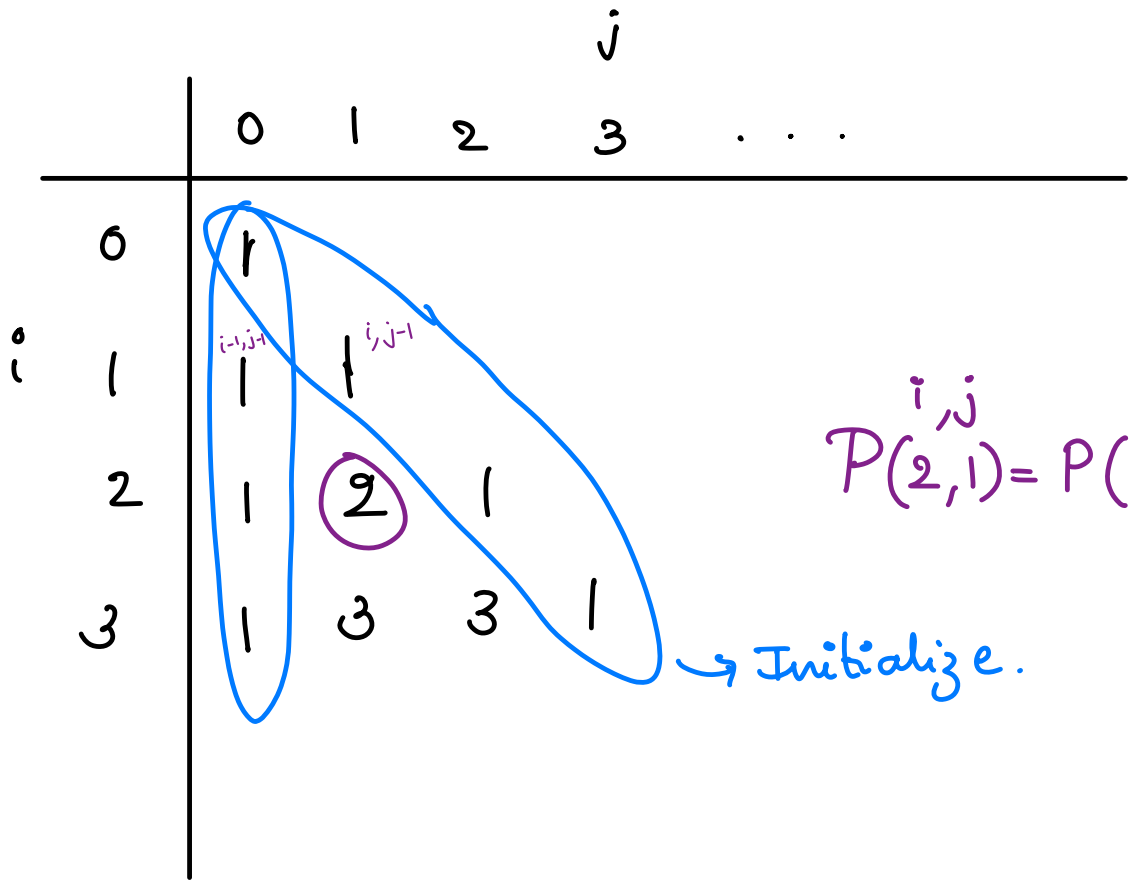
Pascal's Triangle:



$$P(2,1) = P(1,0) + P(1,1)$$

(i, j) (i-1, j-1) (i-1, j)

$$P(i,j) = P(i-1,j) + P(i-1,j-1)$$

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Algorithm:

For $i = 0$ to $n-1$

$B(i,0) \leftarrow 1$; $B(i,i) \leftarrow 1$

For $i \leftarrow 2$ to $n-1$

For $j \leftarrow 1$ to $i-1$

$B(i,j) \leftarrow B(i-1,j) + B(i-1,j-1)$

Time Complexity

$$\sum_{i=2}^{n-1} \sum_{j=1}^{i-1} 1$$

$$= \sum_{i=2}^{n-1} (i-1) - 1 + 1$$

$$= \sum_{i=2}^{n-1} i - 1$$

$$= 1 + 2 + \cdots + n-2$$

$$= \frac{(n-2)(n-1)}{2} \in O(n^2) \;//$$

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Longest Common Subsequence (LCS):

Given a sequence $X = \langle x_1, x_2, \cdots, x_n \rangle$, $Z = \langle z_1, z_2, \cdots, z_k \rangle$ is a subsequence if there exists strictly increasing sequence of indices $\langle i_1, i_2, \cdots, i_k \rangle$ of X s.t $x_{i_j} = z_j$ $\forall$ all $1 \leq j \leq k$.

eg: $X = \langle A, B, C, B, D, A, B \rangle$ ($x_1\ x_2\ x_3\ x_4\ x_5\ x_6\ x_7$)

Is $Z = \langle B, C, D, B \rangle$ ($z_1\ z_2\ z_3\ z_4$) a subsequence of X ?

Strictly increasing indices ($i_1\ i_2\ i_3\ i_4$) $\langle 2, 3, 5, 7 \rangle$

$$X_2 = X_{i_1} = Z_1 = B$$
$$X_3 = X_{i_2} = Z_2 = C$$
$$X_5 = X_{i_3} = Z_3 = D$$
$$X_7 = X_{i_4} = Z_4 = B$$

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$\therefore$ Z is a subsequence of X. //

Note: $\langle D, C \rangle$ is not a subsequence of X.

Longest Common Subsequence.

eg: $X = \langle A, B, C, B, D, A, B \rangle$

$Y = \langle B, C, D, B, A \rangle$.

Longest Common Subsequence of X & Y

$LCS(X, Y) = \langle B, C, D, B \rangle$.

Length = 4.

Brute Force:

$X = \langle x_1, x_2, \ldots, x_m \rangle$

$Y = \langle y_1, y_2, \ldots, y_n \rangle$

No of subsequences

2^m

Compare each with Y.

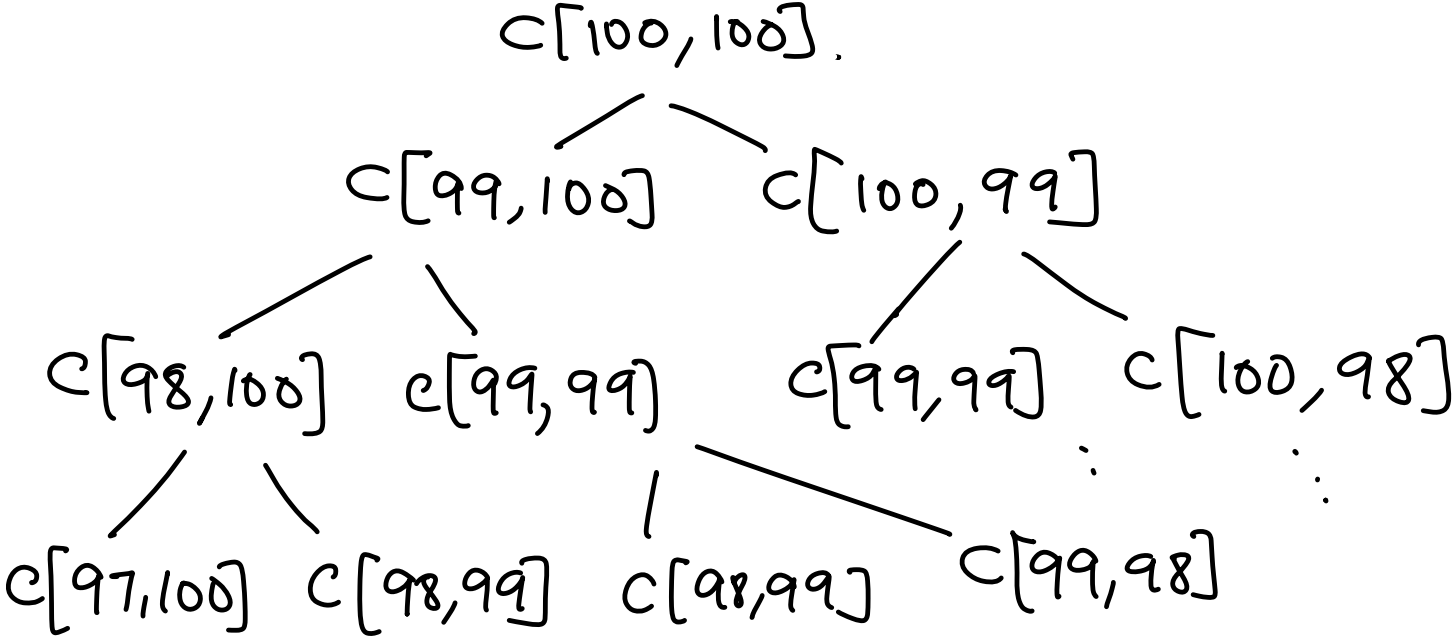
Comparing one subsequence of X with Y takes $O(n)$ time

No of subsequences $= 2^m$

Total time $= O(n \cdot 2^m)$

Recursive LCS:

```
LCS(X, Y, i, j)
  If  X[i] = Y[j]
    then  c[i,j] ← LCS(X, Y, i-1, j-1) + 1.
  else
        c[i,j] ← max{ LCS(X, Y, i-1, j),
                      LCS(X, Y, i, j-1) }
  return  c[i,j].
```



Repetitive Call

(Pattern similar to Fibonacci Recursive Algorithm)

Time Complexity

Let $i+j = \text{size} = N$

$i-1+j = N-1$, $i+j-1 = N-1$

$i-1+j-1 = N-2$

$T(N) = T(N-1) + T(N-1)$ (else case)

$T(N) = T(N-1) + T(N-2)$ ← Fibonacci. (Alternative If & else satisfied.)

$T(N) = T(N-1) + T(N-1)$

$= 2T(N-1)$

$= 2(2T(N-2))$

$= 2^2(2T(N-3))$

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$= 2^3 T(N-3)$

$= 2^k T(1)$

$= 2^{N-1} \in O(2^{N-1})$

$N - k = 1$

$k = N-1$

(Exponential)

Recursive LCS is exponential algorithm.

Dynamic Programming

LCS - length (X, Y):

1. $m \leftarrow$ length [X]
2. $n \leftarrow$ length [Y]
3. For $i \leftarrow 0$ to m
4. $\quad c[i, 0] \leftarrow 0$
5. For $j \leftarrow 0$ to n
6. $\quad c[0, j] \leftarrow 0$
7. For $i \leftarrow 1$ to m
8. $\quad$ For $j \leftarrow 1$ to n

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9. If $x_i = y_j$

10. then $c[i,j] \leftarrow c[i-1,j-1]+1$

11. $b[i,j] \leftarrow$ "↖" (diagonal)

12. else if $c[i-1,j] \geq c[i,j-1]$

13. then $c[i,j] \leftarrow c[i-1,j]$

14. $b[i,j] \leftarrow$ "↑" (top)

15. else $c[i,j] \leftarrow c[i,j-1]$

16. $b[i,j] \leftarrow$ "←" (left)

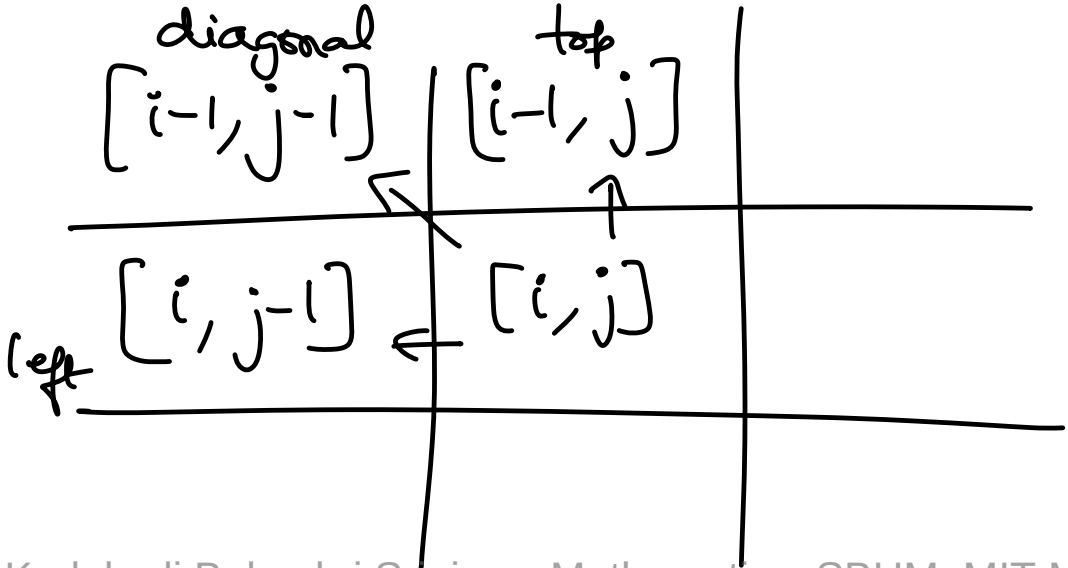
17. return C and b.

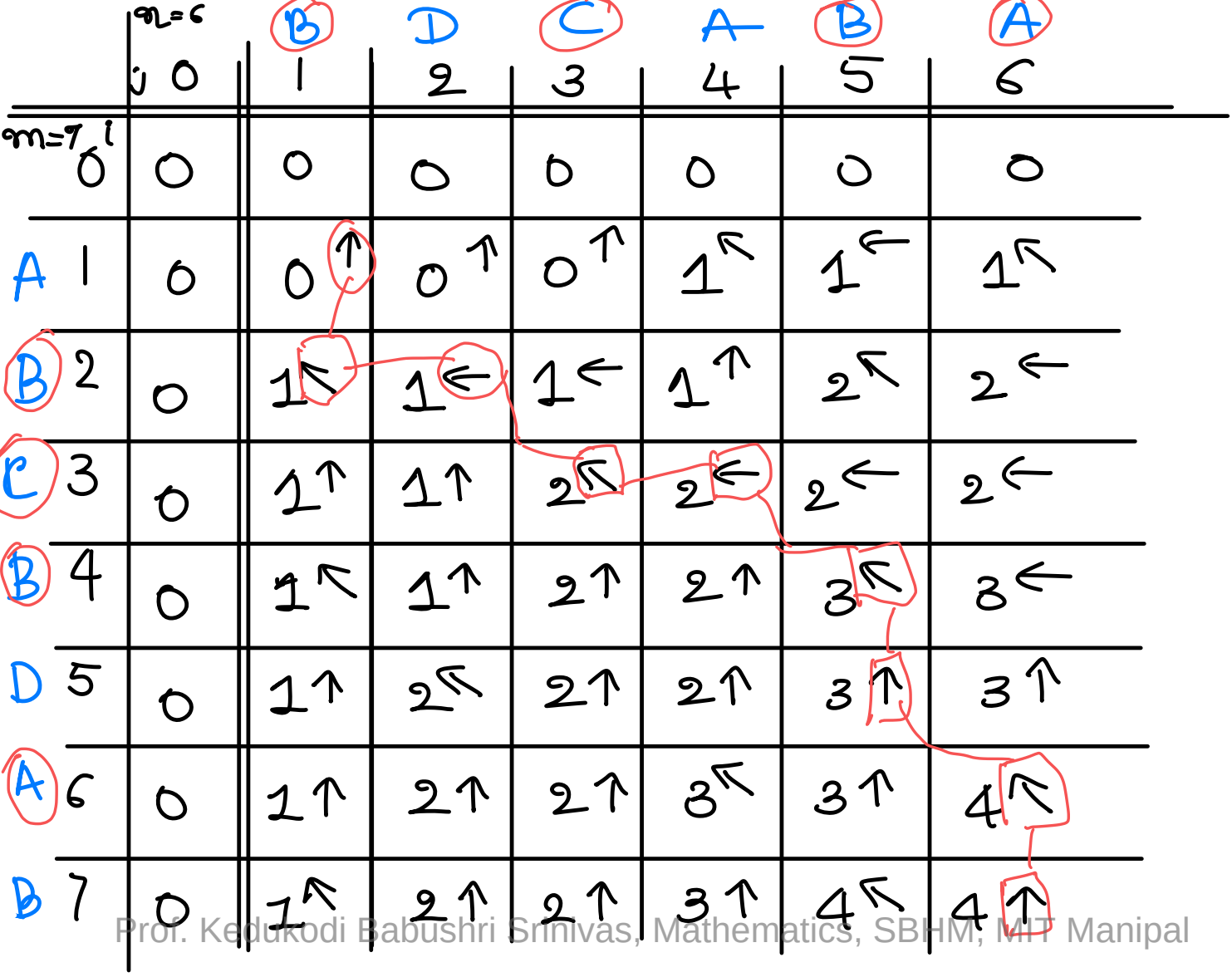
eg:

$X = \langle A, B, C, B, D, A, B \rangle$.

$Y = \langle B, D, C, A, B, A \rangle$

$m = 7 \qquad n = 6$.





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LCS (X, Y)

= reverse of (ABCB)

= BCBA

Length LCS (X, Y) = 4.